Sq

Sigmoid function $\sigma(x) = \frac{1}{1+e^{-x}}$

derivative of sigmoid function

$$\sigma'(x) = \frac{d}{dx}\left(\frac{1}{1+e^{-x}}\right)$$

$$= \frac{d}{dx}\left((1+e^{-x})^{-1}\right)$$

$$= -\frac{1}{(1+e^{-x})^2} \cdot \frac{d}{dx}(1+e^{-x})$$

$$= -\frac{1}{(1+e^{-x})^2} \cdot -e^{-x}$$

$$= \frac{e^{-x}}{(1+e^{-x})^2}$$

$$= \frac{1}{(1+e^{-x})} \cdot \frac{e^{-x}}{(1+e^{-x})} \qquad \left[\sigma(x) = \frac{1}{1+e^{-x}}\right]$$

$$= \sigma(x) \cdot \frac{e^{-x}}{1+e^{-x}}$$

$$= \sigma(x) \cdot \frac{1+e^{-x}-1}{1+e^{-x}}$$

$$= \sigma(x) \cdot \left(\frac{1+e^{-x}}{1+e^{-x}} - \frac{1}{1+e^{-x}}\right) \qquad \left[\sigma(x) = \frac{1}{1+e^{-x}}\right]$$

$$\sigma'(x) = \sigma(x) \cdot \left(1 - \sigma(x)\right)$$

$$\therefore \sigma'(x) = \sigma(x) \cdot \left(1 - \sigma(x)\right)$$